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Western Australia Mineral Exploration RAG Assistant

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- **The Problem:** Geological WAMEX reports are massive, unstructured, and difficult to query at scale.
- **The Solution:** An AI-powered assistant that provides "grounded" answers with direct citations to original PDF sources.
- **The Benefit:** Reduces "Time to Insight" for exploration geologists from hours to seconds.

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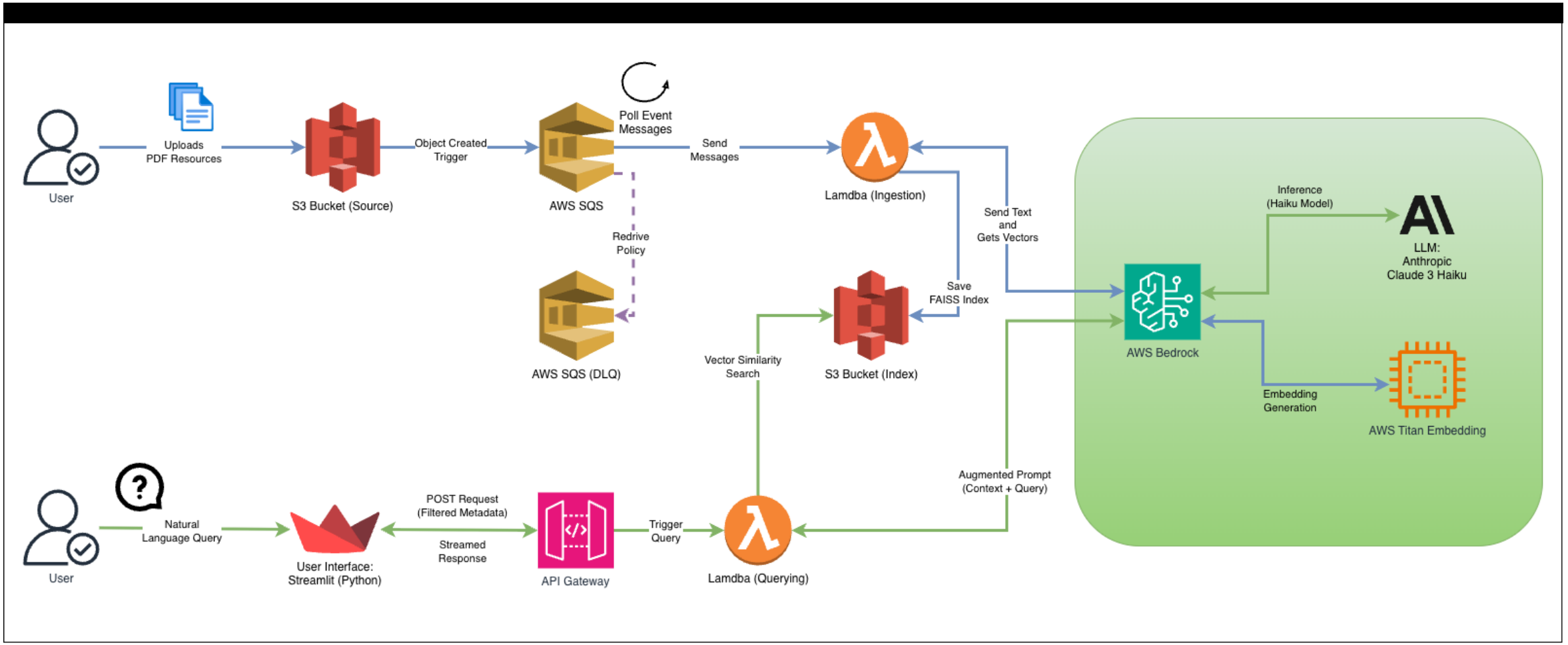

Serverless GenAI: 100% Event-Driven RAG

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Resilient Ingestion: Handling Throttling & High Performance

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns

Actions ▼ | Start tailing | Create metric filter

Filter events - press enter to search | Clear | 1m | 30m | 1h | 12h | Custom | UTC timezone ▼ | Display ▼

Throttling

Timestamp	Message
	No older events at this moment. *Retry*
2026-03-08T00:24:20.769Z	INIT_START Runtime Version: python:3.11.mainlinev2.v3 Runtime Version ARN: arn:aws:lambda:[redacted]
2026-03-08T00:24:23.799Z	[INFO] 2026-03-08T00:24:23.798Z Found credentials in environment variables.
2026-03-08T00:24:23.996Z	START RequestId:[redacted] Version: $LATEST
2026-03-08T00:24:37.871Z	[ERROR] 2026-03-08T00:24:37.870Z [redacted] Error raised by inference endpoint Traceback (most recent call last): File "/var/task/langchain_aws/embeddings/bedroc…
2026-03-08T00:24:37.871Z	[ERROR] 2026-03-08T00:24:37.871Z [redacted] Error processing A143696_v1_REPORT.pdf: An error occurred (ThrottlingException) when calling the InvokeModel operatio…
2026-03-08T00:24:37.872Z	[ERROR] ThrottlingException: An error occurred (ThrottlingException) when calling the InvokeModel operation (reached max retries: 4): Too many requests, please wait before trying again. T…
2026-03-08T00:24:37.874Z	END RequestId:[redacted]
2026-03-08T00:24:37.874Z	REPORT RequestId:[redacted] Duration: 13878.24 ms Billed Duration: 17102 ms Memory Size: 1024 MB Max Memory Used: 193 MB Init Duration: 3222.86 ms
	No newer events at this moment. *Auto retry paused. Resume*

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns

Actions ▼ | Start tailing | Create metric filter

Filter events - press enter to search | Clear | 1m | 30m | 1h | 12h | Custom | UTC timezone ▼ | Display ▼

Success Retries

Timestamp	Message
	No older events at this moment. *Retry*
2026-03-08T00:39:20.681Z	INIT_START Runtime Version: python:3.11.mainlinev2.v3 Runtime Version ARN: arn:aws:lambda:[redacted]
2026-03-08T00:39:23.087Z	[INFO] 2026-03-08T00:39:23.087Z Found credentials in environment variables.
2026-03-08T00:39:23.238Z	START RequestId:[redacted] Version: $LATEST
2026-03-08T00:39:25.924Z	[INFO] 2026-03-08T00:39:25.924Z [redacted] Loading faiss.
2026-03-08T00:39:26.129Z	[INFO] 2026-03-08T00:39:26.128Z [redacted] Successfully loaded faiss.
2026-03-08T00:39:26.359Z	[INFO] 2026-03-08T00:39:26.358Z [redacted] Successfully merged new data into existing FAISS index.
2026-03-08T00:39:26.447Z	[INFO] 2026-03-08T00:39:26.447Z [redacted] Processing A143696_v1_REPORT.pdf from wamex-raw-pdfs [redacted]
2026-03-08T00:39:26.449Z	END RequestId:[redacted]
2026-03-08T00:39:26.450Z	REPORT RequestId:[redacted] Duration: 3210.93 ms Billed Duration: 5765 ms Memory Size: 1024 MB Max Memory Used: 211 MB Init Duration: 2554.06 ms
	No newer events at this moment. *Auto retry paused. Resume*

Optimized for FinOps: $0.00 Idle Cost Architecture

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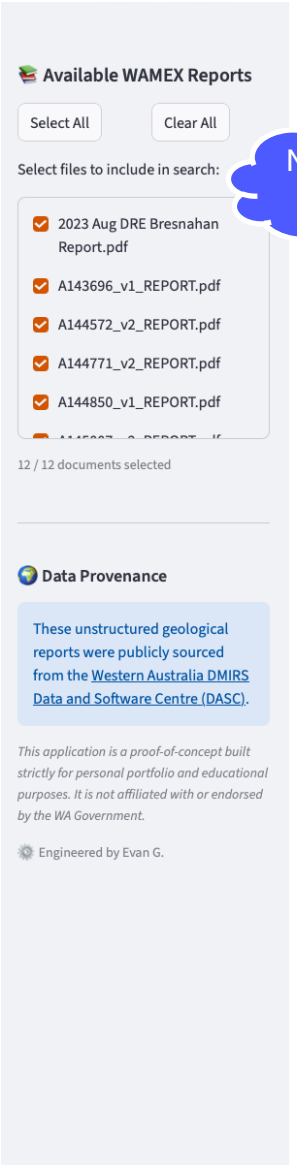
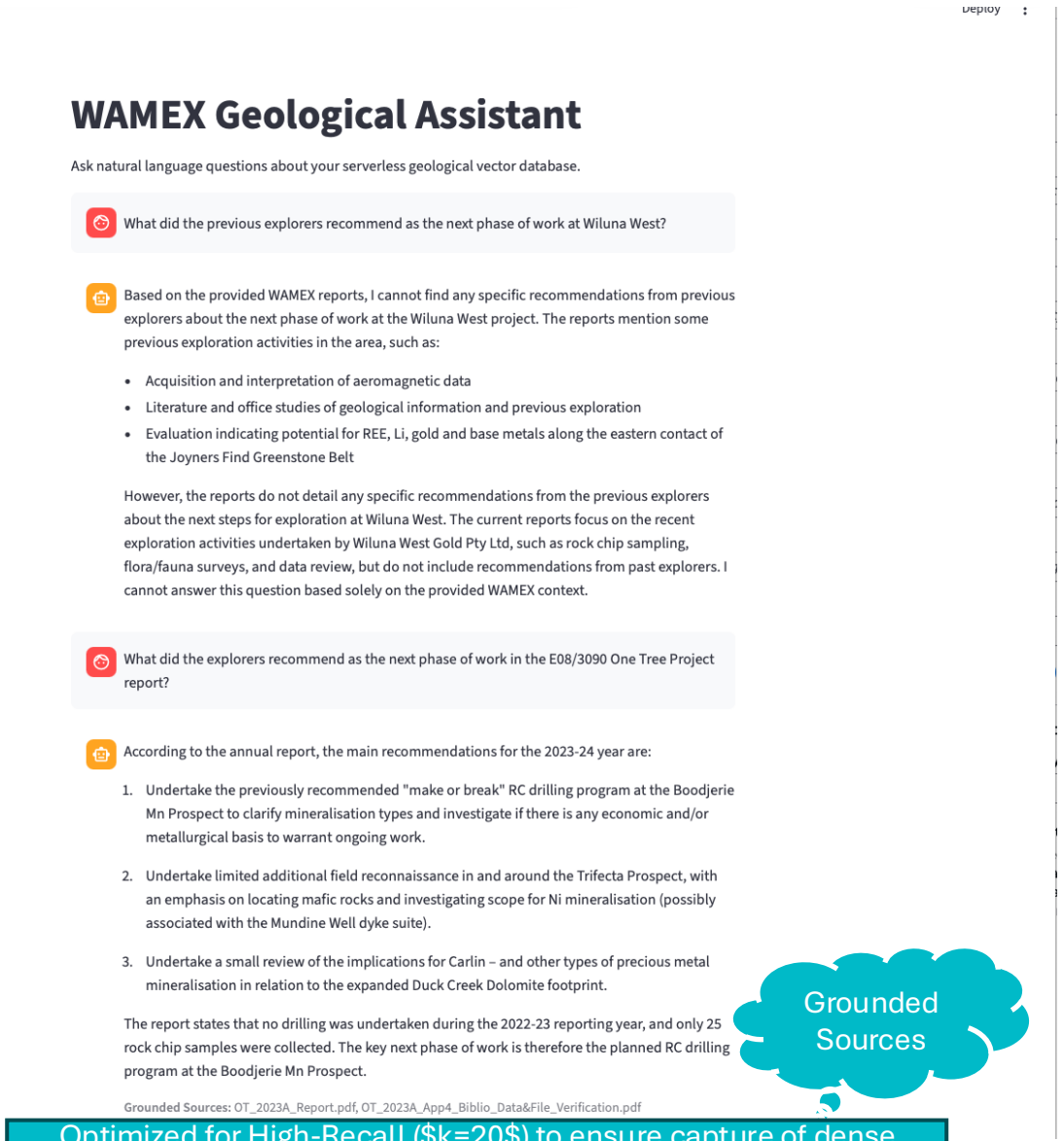
The Final UI: Grounded Geological Insights



Metadata Filtering



Grounded Sources

Optimized for High-Recall ($k=20$) to ensure capture of dense assay results across document segments.

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Outcome & Results

Financial Efficiency (FinOps): Achieved a true $0.00 idle cost architecture by utilizing S3-backed FAISS instead of always-on managed vector clusters.

System Resiliency: Successfully implemented an asynchronous SQS redrive policy that self-heals from ThrottlingExceptions during high-volume ingestion.

Performance Optimization: Document processing (chunking to indexing) completes in < 5 seconds with a minimal memory footprint of ~211 MB.

Search Precision: Fine-tuned retrieval to *k=20* with a 150-token overlap to ensure high-fidelity extraction of dense geological assay data.

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Links:

- Western Australia DMIRS Data and Software Centre (DASC): https://dasc.dmirs.wa.gov.au/
- Github: https://github.com/evan-gloria/wamex-rag-assistant

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